

Exponential Equations

Elex

$$e^{2x} - 5e^x + 6 = 0$$

$$\text{Let } t = e^x \Rightarrow t^2 - 5t + 6 = 0$$

$$\boxed{t_1 = 3, t_2 = 2}$$

then:

For t_1 :

$$e^x = 3$$

$$\ln(e^x) = \ln(3)$$

$$x = \ln(3)$$

For t_2 :

$$e^x = 2$$

$$\ln(e^x) = \ln(2)$$

$$x = \ln(2)$$

Elex